Digital Twin Anchored Generation: Adaptive Querying and Phenotypic Refinement

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Abstract—Digital Twin Anchored Generation (DTAG) constructs a survey-grounded *predictive phenotype* from Large Science Models (LSMs) of high-dimensional response dependencies. Each observed answer refines predictions for unasked items, while an adaptive policy selects questions by their expected displacement of a pole-relative geometric index. Native GSS 2024 experiments with six pinned personas produce 960 adaptive 20-question trajectories and 2,880 fixed-set order replays. Direction-seeking selection yields reproducible displacement toward either reference pole; sets frozen after adaptive selection retain this movement under reversed and shuffled interrogation, without a detected systematic mean order effect. These results characterize how learned conditional structure can be navigated through selective questioning. We formulate a prospective randomized human study to test the transfer of these predictions to real responses and to examine whether progressive questioning also influences answers beyond selective phenotype refinement.

Index Terms—Predictive phenotypes, large science models, conditional response modeling, digital twins, survey methodology, opinion dynamics, adaptive questioning, generative simulation.

1 Introduction

Opinions about social, political, religious, and economic questions form a coupled system: knowing a respondent's answers to a few items changes the expected answers to many others. This structure raises a quantitative question. Can questions be selected to refine a respondent's inferred position in a predictable direction, and how much of the resulting movement follows from measurement selection rather than changes in the respondent?

We introduce Digital Twin Anchored Generation (DTAG), coupling a language-based questioning interface to a Large Science Model (LSM) of empirical conditional survey responses. Observed answers anchor a *predictive phenotype*: a distribution over responses not yet observed. An LSM-derived geometry relates this predictive state to reference configurations, and an adaptive policy selects successive questions by their expected geometric displacement.

Here we develop the framework and evaluate direction-seeking questioning in native GSS 2024 simulations. Fixed-question-set replays separate the effect of selecting particular items from the effect of continuously reoptimizing their sequence. A prospective human experiment then tests whether this conditional geometry predicts directional movement from actual answers. Progressive elicitation may additionally affect the considerations or beliefs available to respondents, a possibility examined through measurements beyond the selected questions.

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2 Prior Work and the Scientific Gap

2.1 Belief systems, cognitive organization, and survey response

Converse [1] identified weak issue constraint outside politically sophisticated groups; Zaller [2] modeled reports as samples from accessible considerations. Later studies separate partisan sorting from issue alignment [3] and show how homophily and influence couple otherwise weak affinities [4]. These motivate measuring conditional attitude organization rather than assuming a single axis.

Work on cognitive consistency [5], survey wording and order [6], and motivated evaluation [7] shows that elicitation can shape which considerations become salient and how attitudes are expressed. These findings motivate distinguishing the refinement of an inferred phenotype from possible effects of questioning on later reports.

2.2 Quantitative influence and experimental evidence

DeGroot [8], Friedkin–Johnsen [9], and bounded-confidence [10] models specify social transitions but typically use low-dimensional individual states. DTAG instead models high-dimensional conditional structure without yet identifying a causal human transition operator.

Bail *et al.* [11] randomized incentives to follow Twitter bots providing cross-cutting political content. Republicans assigned a liberal bot later reported more conservative views; the corresponding Democratic estimate was not significant. These observations motivate examining how heterogeneity interacts with structured questioning. Campaign-contact experiments found small general-election mean effects [12], whereas some structured conversations produced more persistent reported changes [13]. These studies motivate examining heterogeneity in responses to structured sequences and distinguishing immediate response effects from persistent attitude change.

2.3 Generative simulation and conditional response prediction

Generative agents and profile-conditioned language-model respondents can produce survey-like behavior [14], [15]. DTAG instead grounds generated answers in empirically estimated

dependencies among survey variables. Its conditional distributions describe the range and likelihood of unasked responses given observed answers, while explicit anchors and state updates make the refinement process inspectable and reproducible.

3 Large Science Models and the Predictive Phenotype

3.1 Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ range over $\mathcal{X} = \prod_i \mathcal{A}_i$, with finite response alphabets $\mathcal{A}_i$ and survey population, wave, or context c. An LSM is the conditional family

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated using coupled conditional inference trees [16]–[18]. Edges $j \to i$ indicate prediction, not causation; separately fitted conditionals may be jointly incompatible [19].

The respondent state is a partial assignment $s = \{X_j = a_j : j \in S\}$. In $\widetilde{x}_{-v}(s)$, all unanchored coordinates remain missing (no modal completion); the native trees marginalize missing splits by subtree-mass-weighted branch mixtures. This is a tree-wise prediction rule, not necessarily marginalization under one coherent joint law. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

3.2 Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{4}$$

Here D_{JS} is Jensen–Shannon divergence as implemented by the native predictive-distance routine. Predictively indistinguishable states may be at zero distance: this is generally a pseudometric of *predictive consequence*, not a postulated Euclidean manifold of minds.

With 80% responses masked [18], mean LSM-distance distortion declined about 54.9% (GSS) and 60.9% (Eurobarometer). These results quantify the fidelity with which incomplete observations recover the model's predictive geometry.

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{5}$$

The semantically defined poles [18] specify 27 opposing GSS 2024 responses on social policy, civil liberties, public spending, and religion (Table 1); all other coordinates remain unspecified. They are reference configurations, not typical respondents. The pole-relative index is not a self-reported ideology score.

4 DTAG: Conditional Response Generation and Adaptive Querying

4.1 Architecture and generative semantics

A semantic interface maps question q to survey variables $M(q)$. The LSM predicts responses, the answer anchors the state, and an LLM verbalizes it. For a selected survey item, the response is sampled from the conditional LSM distribution:

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \qquad v_t \in M(q_t), \tag{6}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{7}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{8}$$

The sampled response is added to the anchored state and the predictions for unasked items are updated; initial persona anchors remain fixed. The optimizer selects questions using expected displacement but does not select the respondent's answers.

Figure 1 combines the architecture and native experiment.

4.2 Sequential geometry and adaptive questioning

For every candidate item v and response a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{9}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v) \Delta_v(a; s), \tag{10}$$

$$\sigma_v^2(s) = \sum_{a} p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{11}$$

The expected realized displacement equals the selection score:

$$\mathbb{E}[I_\Phi(s_{t+1}) - I_\Phi(s_t) \mid s_t, v_t] = \mu_{v_t}(s_t).$$

This identity specifies the local optimization objective; the attainable movement over 20 questions, variation across initial states, and retention after sets are frozen are empirical properties of the fitted conditional system. The reported policies evaluate this objective under sampled conditional responses.

The predictive change at unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \tag{12}$$

A weighted score $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$, with prespecified $w_j \geq 0$, quantifies how one answer redistributes predictions across the remaining variables.

4.3 Difference from RAG and language-model personas

RAG provides grounding text [20]; profile-prompted LLM personas seek plausible or distributionally faithful responses [14], [15]. DTAG instead exposes a native conditional model whose predictions for unasked responses can be calibrated as anchors accumulate. It complements text retrieval through an explicit probabilistic response mechanism. The public DTAG implementation includes the native inference engine and a browser-based workbench (https://github.com/zeroknowledgediscovery/DTAG/tree/main/webapp).

TABLE 1

GSS 2024 left/right reference poles. The 27 listed item–response assignments are the complete nonmissing coordinates of the two pole states used to compute I_Φ. All other survey variables are unspecified. Topics paraphrase item wording; categorical responses match the simulation definitions. The poles yield $I_\Phi = -1$ and $+1$.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove

Item	Survey topic	Left	Right
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

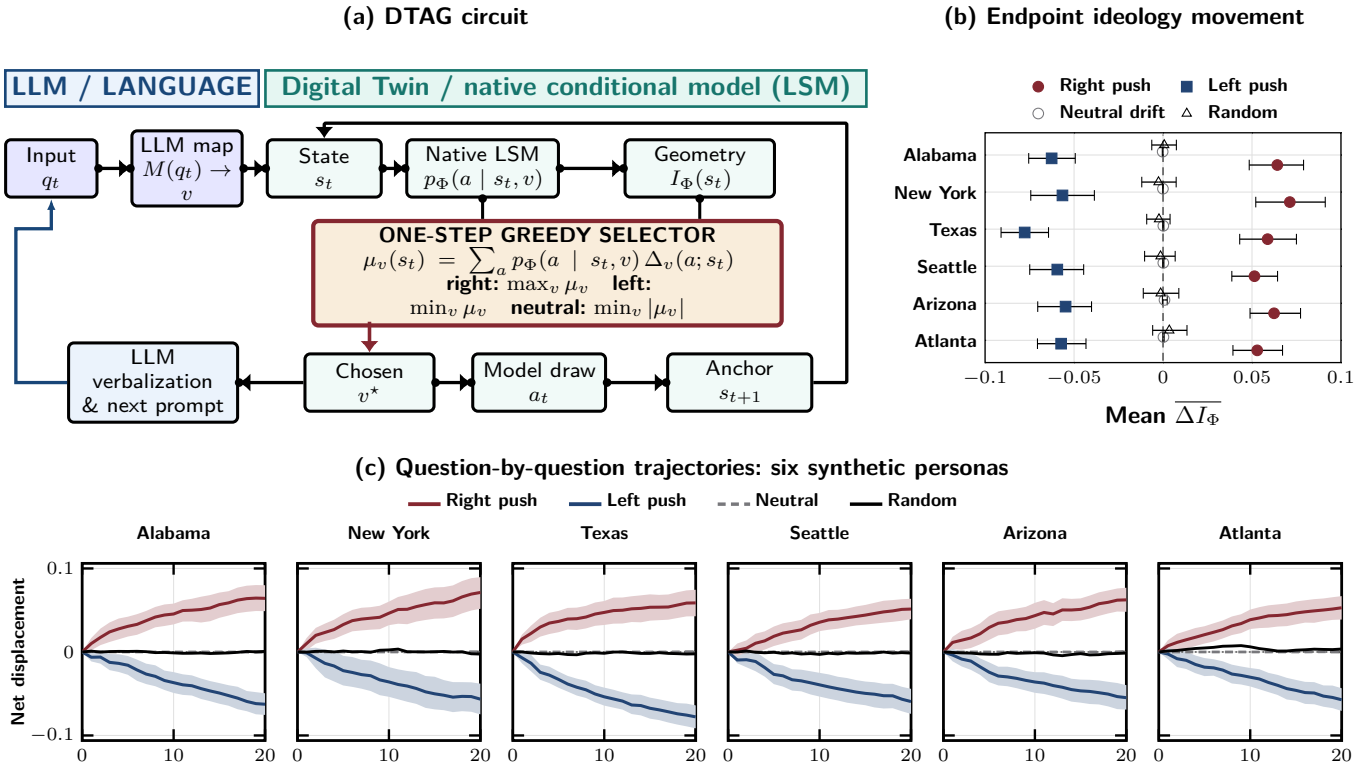


Fig. 1. **DTAG digital-twin circuit and adaptive native GSS 2024 results.** (a) A semantic LLM interface maps and verbalizes categorical survey questions, while the conditional LSM inside the digital twin predicts unasked responses; a myopic greedy circuit selects the next question using expected one-step pole-relative displacement and updates the anchored state with a model draw. (b) Mean displacement after 20 questions across six synthetic personas; large markers and horizontal whiskers show 95% bootstrap uncertainty. (c) Six aligned trajectories from 40 repetitions per persona and policy; shading denotes pointwise 95% bootstrap intervals. Right = dark red, left = dark blue, neutral = dashed gray, random = black. The trajectories show model-generated phenotype refinement under each selection policy.

5 The Interpretation of Sequential Questions

5.1 Phenotypic refinement: observing a fixed system

Under passive refinement, respondents possess fixed phenotypes $X^\star$ and answers increasingly constrain predictions along $s_0 \subset s_1 \subset \cdots \subset s_T$. For target items Y, the information value of query v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v|s}[H(Y \mid s, v = A_v)]. \quad (13)$$

For a compatible joint law this is nonnegative conditional mutual information, distinct from geometric displacement μ_v in (10). Held-out response prediction as the observed answer set grows provides a direct assessment of phenotype refinement; fixed archival response records offer one possible external evaluation.

5.2 Phenotypic construction: querying a reflexive system

Questioning may also affect subsequent reports or attitudes. To represent this possibility, let Z_t denote a respondent's latent state and define the response–transition operator

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \quad (14)$$

TABLE 2
Six pinned synthetic personas in the native GSS 2024 simulations.

Persona	Pinned description	Initial $I_\Phi(s_0)$
Alabama	45-year-old white man living in open country outside a small town in rural Alabama, in the South. He is married and has two children. He works full time in farming. He is a veteran who served 2 to 4 years of active duty in the armed forces. He reads a newspaper every day. His education ended with a high school diploma. He thinks of himself as conservative and is a strong Republican. He is Protestant and attends religious services every week.	0.075706
New York	22-year-old white woman living in New York City, a large central city in the Northeast. She has never been married and has no children. She works full time in retail. She has never served in the armed forces. She reads a newspaper every day. She has a bachelor's degree. She thinks of herself as extremely liberal and is a strong Democrat. She has no religion and never attends religious services.	−0.163807
Texas	61-year-old white woman in a small town in Texas, in the South. She is married with three children and keeps house. Her education ended with high school. She thinks of herself as extremely conservative and is a strong Republican. She is Protestant and attends religious services several times a week. She reads a newspaper every day.	0.045379
Seattle	34-year-old white man living in Seattle, a large central city in the West. He has never been married and has no children. He works full time and has a graduate degree. He thinks of himself as liberal and is a Democrat, but not a very strong one. He has no religion and never attends religious services. He reads a newspaper a few times a week.	−0.119639
Arizona	39-year-old Mexican American man living in a medium-size city in Arizona, in the West. He is married with three children and works full time. His education ended with high school. He thinks of himself as conservative and is a Republican, but not a very strong one. He is Catholic and attends religious services nearly every week.	−0.020437
Atlanta	47-year-old Black woman living in Atlanta, a large central city in the South. She is divorced with two children and works full time. She has a bachelor's degree. She thinks of herself as liberal and is a strong Democrat. She is Protestant and attends religious services every week. She reads a newspaper every day.	−0.030807

Persona descriptions and initial pole-relative ideology index $I_\Phi(s_0)$. Positive values point toward the designated right pole and negative values toward the left pole.

Here O_q generates the response from the current state, and T_q describes any change following the question and answer. Passive observation corresponds to $T_q(z' \mid z, a) = \delta_z(z')$; nontrivial transitions may reflect altered accessibility, interpretation, or belief. The LSM describes conditional reports, while a human transition operator requires experimental identification.

A common held-out battery following randomized questioning can test whether earlier question selection affects later reports beyond selective revelation. Delayed measurement can distinguish transient response-context effects from more persistent changes, potentially including persuasion.

5.3 Question order and conditional compatibility

For fixed realized answers, the terminal partial state and its computed I_Φ do not depend on the order in which answers were entered. The *distribution* of sequentially generated answers may nevertheless be order-dependent when fitted conditionals are incompatible. A coherent static joint distribution satisfies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \tag{15}$$

Separately fitted conditional models need not satisfy this equality [19]. In human respondents, question-dependent K_q operators introduce a further possible source of order effects. Fixed-set native replay evaluates the model's order sensitivity while holding the selected questions constant.

6 Native GSS 2024 Computational Experiments

6.1 Adaptive questioning from pinned persona states

We initialized six synthetic respondents from the pinned GSS 2024 states in Table 2. Each repetition draws a new response trajectory conditioned on the same initial anchors; these are independent synthetic response realizations, rather than repeated observations of one fully specified person. We queried survey items directly without executing the LLM mapping or verbalization layer. The native optimizer maximized expected rightward drift, minimized it for leftward drift, minimized absolute drift (neutral), or sampled eligible items randomly. For every policy, the eligible item pool excluded *all 27 variables used to define the left and right reference poles* (Table 1); no pole-defining item was asked in any experimental trajectory. Answers to selected non-pole items were sampled from the LSM conditionals. At $T = 20$ and 40 repetitions per persona/policy, $6 \times 4 \times 40 = 960$ trajectories contain 19,200 sampled answers.

Across all six initial states, the right- and left-seeking policies yielded displacement in their respective directions. Because the reference-pole items were excluded, these displacements arise from changes in the inferred phenotype when non-pole responses are observed, rather than from directly querying the items defining the index. The random policy provides a selection baseline; the neutral policy tests whether low expected geometric displacement can be maintained as the observed state grows.

6.2 Question selection as observed paths

Question identities and policy coverage are summarized in Table 3. Each observed item was assigned a shared categorical index without reordering runs. The 18 displayed right/left/neutral heatmaps contain 161 distinct items across six personas and 40 repetitions per policy. The global index permutation approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \left|\pi(v_{purt}) - \pi(v_{pur't})\right|, \tag{16}$$

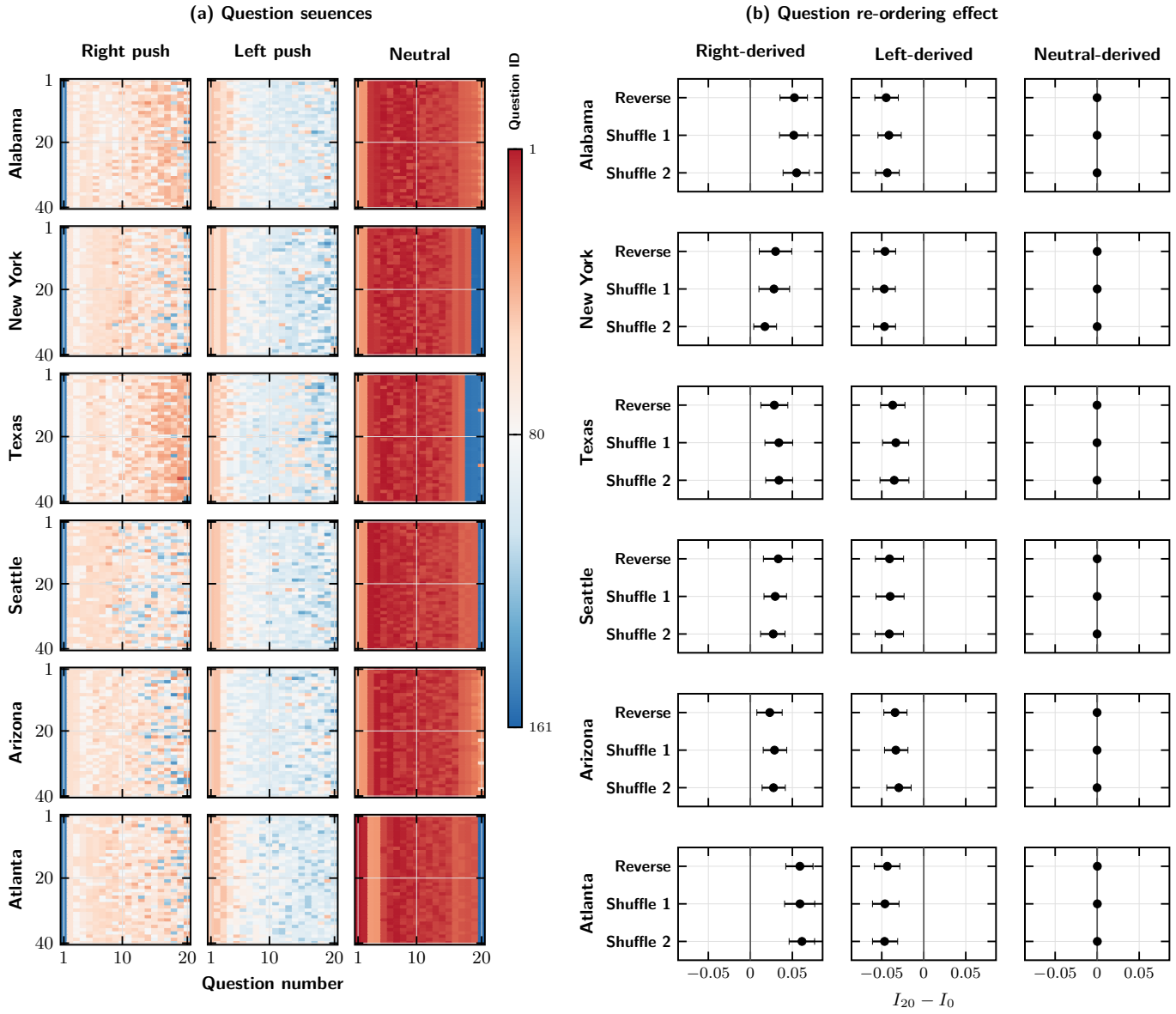


Fig. 2. **Question identities and fixed-item directional displacement across six personas.** Left: 18 categorical heatmaps show 40 runs over 20 question positions, with a shared colorbar identifying questions rather than ideological direction. Right: baseline-relative displacement $I_{20} - I_0$ and 95% bootstrap intervals for fixed optimized question sets replayed in reverse and two shuffled orders (40 repetitions each). Right- and left-derived sets retain directional movement; neutral-derived sets remain near zero. These are fixed-set replays, not the fully adaptive runs in Fig. 1.

TABLE 3
Question selection in the native GSS 2024 experiment.

Selection policy	Distinct	Selections	Pole overlap
Right-seeking	100	4,800	0
Left-seeking	75	4,800	0
Neutral	43	4,800	0
Three-policy union	161	14,400	0
All four policies	228	19,200	0

The random policy contributed 4,800 selections; 67 of the 228 distinct items occurred only under random selection.

Here u ranges over displayed policies. The all-pairs objective is independent of run numbering. The 67 random-only items remain in the lookup; index distances have no scientific meaning.

6.3 Fixed-set replay: robustness to question ordering

For each of six personas and three source policies (right, left, neutral), we froze the 20 questions from adaptive replicate zero without selecting on outcomes. Each set was replayed in its original, reversed, and two shuffled orders, with 40 fresh conditional draws per order ($18 \times 4 \times 40 = 2{,}880$ runs). Question-specific uniform draws were coupled across orders to reduce sampling variability.

Figure 2 shows baseline-relative displacement for the reordered fixed sets. Directional movement persists: averaged across the six persona means, right-derived sets yield +0.0381 (original) and +0.0377 (reversed), while left-derived sets yield −0.0399 and −0.0410. For both original and reversed replays, all six persona-specific pointwise 95% bootstrap intervals exclude zero in the intended direction; both shuffles give the same directional pattern, and neutral-derived drift remains

negligible.

All 54 pointwise bootstrap intervals for *paired mean between-order contrasts* include zero; no systematic mean endpoint difference was detected among the tested permutations. The selected question sets therefore retain directional effects without continued reoptimization. In conjunction with the adaptive results in Fig. 1, these replays distinguish question-set effects from sequential reselection and tested order effects.

6.3.0.1 Cross-environment limitation.: Historical ten-question local and CI runs yielded different initial indices despite matching persona, semantic-map, and pole-file hashes. Model binaries were not hashed in both manifests, so absolute cross-environment comparison requires runtime/model parity checks. Within-study results remain consistent.

7 Testable Hypotheses and Prospective Validation

H1 (adaptive phenotype refinement). From a specified initial state s_0, directional optimization of $\mu_v(s)$ should yield repeatable, policy-specific displacement of the pole-relative index $I_\Phi(s)$ across independently sampled response sequences. The native trajectories and fixed-set replays evaluate this prediction and identify contributions from question selection and order.

H2 (transfer to human responses). With the same initialization protocol, questions chosen by the LSM-based policy should produce directional displacement of I_Φ computed from human answers, relative to neutral and random selection. This is a direct out-of-model test of whether learned conditional geometry supports predictive adaptive questioning.

7.0.0.1 Prospective Prolific experiment.: Respondents will complete a common baseline battery before randomization to right-seeking, left-seeking, neutral, or random question selection. The arms will have a matched-length adaptive phase, and the preregistered primary outcome will be $I_\Phi(s_T) - I_\Phi(s_0)$, using respondents' actual answers. Prespecified directional contrasts between arms will test H2; question-level predictive scores will assess calibration. A common held-out battery, administered after the adaptive phase, will assess whether question history also affects responses not selected by the policy. Such differences would indicate an effect on subsequent reporting beyond selective refinement, while optional delayed measurement would assess persistence. Pilot data will determine study power, with balanced protocols, informed consent, and debriefing. Prolific supports recontact [21].

8 Discussion and Limitations

The native experiments characterize selective refinement within the geometry of an empirically fitted survey-response system. Each answer constrains the remaining predictive phenotype; the pole-relative index tracks the implied position of that changing conditional state. The directed trajectories and fixed-set replays show that adaptive optimization and the selected item sets yield reproducible geometric displacement. Effect magnitudes depend on pole construction, the candidate item pool, initial anchors, and model calibration. The index describes predictive position relative to reference configurations, not a self-reported ideology score.

In human questioning, progressive elicitation may additionally reorganize accessible considerations or alter subsequent judgments. These effects might be transient, or they might persist as attitude change or persuasion. A common post-battery and delayed measurement can separate selective refinement from history-dependent reporting and investigate persistence. Model incompatibility may also create synthetic ordering effects, and the native experiments do not estimate a human belief-transition operator. These questions motivate the proposed external validation and continued sensitivity assessment of the reference poles and conditional model.

9 Conclusion

DTAG turns learned conditional survey geometry into an operational strategy for selecting questions that refine a predictive phenotype in a chosen direction. Native 20-question experiments demonstrate directional trajectories across six synthetic personas; fixed-set replays retain substantial movement under alternative orders and without continued optimization. The proposed human study will evaluate the transfer of this directional prediction to real answers and explore whether progressive questioning has further effects on subsequent reports, including the possibility of persistent attitude change.

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